



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2022-23
MONDAY TEST

CLASS : X
SUBJECT: MATHEMATICS

FULL MARKS: 40
DATE: 17.06.22

General Instructions:

- The paper consists of three printed pages.
- All questions are compulsory.
- Copy the question number carefully before answering the questions.

SECTION: A

1. i) The nature of the roots where discriminant is 0 will be: [5 × 1 = 5]

- a) real and distinct b) non-real c) equal d) real and equal

ii) In an interstate purchasing, the tax applied is called:

- a) GST b) CGST c) IGST d) SGST

iii) The set of numbers from which the value of a variable is chosen is called a:

- (a) solution set (b) replacement set (c) complement set (d) disjoint set

iv) The maturity value of a recurring deposit is [P = amount deposited monthly, I = total interest, n = number of months] :

- a) $Pn + I$ b) Pn c) $P+I$ d) $Pn + In$

v) Which one of the following cannot be reduced to a quadratic equation?

- (a) $(x + 2)^2 = 2(x + 3)$ (b) $x^2 + 3x = (-1)(1 - 3x)^2$ (c) $(x + 2)(x - 1) = 5x^2 - 2x - 3$
(d) $x^3 - x^2 + 2x + 1 = 5$

SECTION: B

2. A manufacturer gives a discount of 20% to a wholesaler on a mobile phone marked at ₹15000. The wholesaler then sells it to a retailer at 10% discount on marked price. If a retailer sells it to the customer at a marked price and GST charged is 12% at each stage of intra state transaction Find the:

(i) Cost price of the wholesaler:

(ii) GST paid by the retailer to the State Govt.

(iii) Price the customer pays for the mobile phone including GST. [4]

3. Prakash deposits ₹ 400 every month in a bank's recurring deposit account for $2\frac{1}{2}$ years. If he gets ₹ 13085 at the time of maturity find:

(i) the total interest earned by Prakash (ii) the rate of interest [4]

4. Find the values of m for which the following equation has equal roots:

$$x^2 + mx + (m + 3) = 0 \quad [4]$$

5.a) Solve the following inequation, write the solution set and represent it on the number line: $\frac{3}{2} > x + \frac{5}{2} > \frac{-3}{2}; \quad x \in R;$

b) Write the solution set and also find the smallest value of x such that:

$$-3(2 + x) < 2(3x - 1); x \in W; \quad [3 + 3 = 6]$$

6. A dealer at Delhi purchases an article from a manufacturer at Delhi at a price of ₹10,000 and sells it to a retailer at Delhi at ₹ 12,000. If the rate of GST is 18%, Find:

i) Total CGST paid by the Manufacturer:

ii) Net SGST paid by the dealer to the Government:

iii) Total GST collected by the Government: [4]

7. Solve the following equation:

$$\frac{x+8}{3x-5} = \frac{x-8}{x+5}; \quad x \neq -5, \frac{5}{3} \quad [4]$$

8. Meera has a recurring deposit account in a bank for 4 years at 10% p. a. simple interest. If she gets ₹ 5880 as interest at the time of maturity, find:

(i) the monthly instalment (ii) the matured amount [4]

9. Mrs Bose went to a departmental store and bought the following items. The discount and GST rates and the quantity of each item and marked price of each are given below:

Item	Quantity	Rate per item(₹)	Discount	GST
Blazer	1	₹ 3,500	10%	18%
Pair of shoes	2	₹1,200	5%	12%
Food grains	3	₹ 500	5%	NIL
Bakery products(unpacked)	4	₹ 200	NIL	NIL

Calculate the:

i) The total amount of GST paid.

ii) The total bill amount paid by Mrs Bose.

[5]